

FILTER DESIGN TEST CASE

Coupled Resonator Filter - Band 5 (850 MHz) for WiFi 6E Coexistence Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P2 - High

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting Band 5 (850 MHz) for WiFi 6E Coexistence Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 3812.3 MHz and bandwidth of 345.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the WiFi 6E Coexistence Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: Band 5 (850 MHz)
- Application: WiFi 6E Coexistence Filter
- Substrate: 128° YX LiNbO3
- Package: LGA (Land Grid Array)
- Design Tool: Ansys HFSS 2024R1
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Coupled Resonator Filter design targeting Band 5 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3812.3 MHz
Bandwidth (3 dB):	345.1 MHz
Insertion Loss Target:	≤ 3.8 dB
Return Loss Target:	≥ 21.0 dB
Out-of-Band Rejection:	≥ 33.0 dB
Isolation (Duplexer):	≥ 18.0 dB
Q Factor:	7416
Temperature Coefficient:	-21.3 ppm/°C
Die Size:	0.6 x 1.5 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Ansys HFSS 2024R1 and load the Coupled Resonator Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 3812.3 MHz, BW 345.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11437 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.8 dB
- Return loss in passband shall be >= 21.0 dB
- Out-of-band rejection at +/- 518 MHz offset >= 33.0 dB
- Group delay variation across passband shall be <= 7 ns
- Temperature drift over -40°C to +85°C shall be <= 10.2 MHz
- Power handling shall be >= +28 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2024-03-28
Devices Tested: 97
Insertion Loss (meas): 3.5 dB
Return Loss (meas): 23.0 dB
Rejection (meas): 35.6 dB
Isolation (meas): 23.0 dB
Q Factor (meas): 7416
Group Delay Variation: 2.6 ns
Power Handling: +27 dBm
Die Yield: 90.5%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated below-target performance for Band 5 (850 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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